LESSON 9.2

Analyze Graphs of Relationships Between Two Variables

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can use a table and a graph to determine and analyze the relationship between two variable quantities.

WHY THIS MATTERS The ticket shows the cost to attend a college football game. How does the total cost of tickets relate to the number of tickets purchased?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
graph *Spanish: gráfica*	A picture on a coordinate plane that shows how two quantities are related.	points that rise in a line
table of values *Spanish: tabla de valores*	A list of matched pairs of values that shows a relationship using numbers.	months 1, 2, 4 beside cost 24.95, 49.90, 99.80 — one row per pair.
ordered pair *Spanish: par ordenado*	Two numbers written as (x, y) that name one point on a graph. The independent variable comes first.	(3, 12)
axis *Spanish: eje*	One of the two number lines that frame a graph. Each axis is labeled with a quantity from the problem.	x-axis and y-axis
coordinates *Spanish: coordenadas*	The numbers in an ordered pair that tell how far to move along each axis to reach a point.	x-coordinate, y-coordinate
Rate of change *Spanish: Tasa de cambio*	How much y changes each time x goes up by one.	Up $4 for each hour.

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

A table for y = 4x has x = 1, 2, 3. Plot the points.

1. Write the pairs: (1, 4), (2, 8), (3, 12).
2. Plot (1, 4): right 1, then up 4. Plot the others the same way.
3. Check: the points line up.

Answer: three points in a straight line

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

The tram travels 1,200 feet every minute. Which quantity is independent — time or distance?

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The independent variable is _____ , so it goes on the _____ axis.
- The ordered pair (_____ , _____) means that _____ .
- To plot a point I move _____ first, then _____ .
- As x increases, y _____ , so the points _____ .

Word bank graph · ordered pair · coordinates · axis · plot · point · table · input · output · rate · increases · origin

WATCH OUT!

A common mistake

Reversing the coordinates of a point

LESSON 9.2

Analyze Graphs of Relationships Between Two Variables

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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LEARNING TARGET I can use a table and a graph to determine and analyze the relationship between two variable quantities.

WHY THIS MATTERS The ticket shows the cost to attend a college football game. How does the total cost of tickets relate to the number of tickets purchased?

1 **Circle the letter of the best answer. Show how you know.**

Football tickets cost $45 each. What is the total cost of 3 tickets?

(A) $45
(B) $48
(C) $90
(D) $135

HINT What is the cost of ONE ticket?

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Which ordered pair belongs on the ticket graph for 2 tickets?

(A) (2, 90)
(B) (2, 45)
(C) (2, 47)
(D) (90, 2)

HINT Which quantity goes first in the ordered pair?

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

On the ticket graph, what should the two axes be labeled?

(A) Number of tickets and price of one ticket
(B) Number of tickets and total cost
(C) Total cost and $45
(D) Number of games and number of tickets

HINT Which TWO quantities change in this situation?

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

The Sandia Peak tram travels 1,200 feet each minute. How far has it traveled after 3 minutes?

(A) 1,200 feet
(B) 1,203 feet
(C) 2,400 feet
(D) 3,600 feet

HINT How far does the tram go in ONE minute?

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

A table shows a train traveling 48 kilometers every hour: 1 hour, 48 km; 5 hours, 240 km. How many kilometers does it travel in 2 hours?

(A) 48 km
(B) 50 km
(C) 96 km
(D) 144 km

HINT How far in ONE hour?

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

As the number of tickets purchased increases, the total cost…

(A) decreases

(B) stays the same

(C) cannot be determined

(D) increases

HINT Compare the totals for 1, 2, and 3 tickets.

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Find a card you called a match and your partner did not. Pick one ordered pair from it and test it in cost = 45 × tickets — what does the point on the graph tell you that the table does not?

SENTENCE STARTER The ordered pair ___ does / does not match because 45 × ___ = ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 9.2

Analyze Graphs of Relationships Between Two Variables

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can use a table and a graph to determine and analyze the relationship between two variable quantities.

WHY THIS MATTERS The ticket shows the cost to attend a college football game. How does the total cost of tickets relate to the number of tickets purchased?

PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

The ticket table and ticket graph are two representations. Which statement about them is true?

(A) The graph shows a different relationship than the table

(B) Only the table can show that cost increases with tickets

(C) Only the graph is a real mathematical representation

(D) Both show the same relationship — the table with numerical values, the graph in a more visual way

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

The Sandia Peak tram travels 1,200 feet each minute and the ascent takes 15 minutes. How long is the whole ascent in feet?

(A) 1,215 feet

(B) 12,000 feet

(C) 16,800 feet

(D) 18,000 feet

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

A tram starts at 0 feet and travels 1,200 feet each minute. On its graph, minutes are on the horizontal axis and feet on the vertical axis. Which ordered pair lies on this graph?

(A) (5, 6000)
(B) (5, 1200)
(C) (5, 5000)
(D) (6000, 5)

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

A table pairs hours with kilometers: (1, 48), (2, 96), (5, 240). How many kilometers are traveled each hour?

(A) 24
(B) 48
(C) 96
(D) 240

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Why is the line on the football ticket graph dashed instead of solid?

(A) The graph is unfinished
(B) Dashed lines always mean the cost is decreasing
(C) The ticket price changes between points
(D) You can only buy whole numbers of tickets, so the points between whole numbers do not make sense

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

Using the ticket table, what are the coordinates of the point for 4 tickets?

(A) (4, 180)
(B) (4, 160)
(C) (4, 49)
(D) (180, 4)

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Find a card you called a match and your partner did not. Pick one ordered pair from it and test it in cost = 45 × tickets — what does the point on the graph tell you that the table does not?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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Analyze Graphs of Relationships Between Two Variables

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can use a table and a graph to determine and analyze the relationship between two variable quantities.

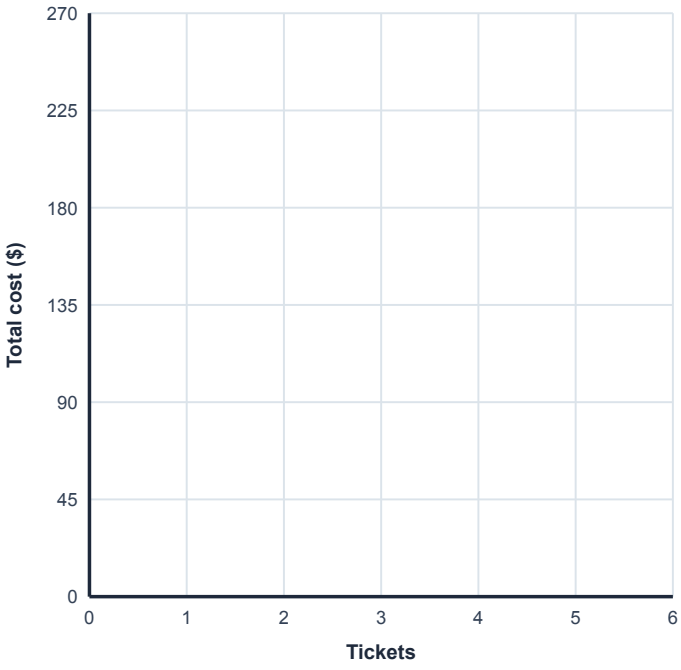
WHY THIS MATTERS The ticket shows the cost to attend a college football game. How does the total cost of tickets relate to the number of tickets purchased?

PART 1 Understand the idea

models and meaning

1 **Plot and label each point on the grid.**

Plot the ticket-cost relationship: one point for each number of tickets.



SHOW YOUR THINKING

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

Nakai's savings account has a principal of $750 and earns 3.5% simple interest each year. How much interest does Nakai earn in one year?

(A) $3.50

(B) $26.25

(C) $75.00

(D) $262.50



0% 50% 100%

WORK

3 **Circle the letter of the best answer. Show how you know.**

Nakai earns $26.25 in simple interest every year. On the graph of years vs. interest earned, which point represents 4 years?

(A) (4, 105)

(B) (4, 26.25)

(C) (4, 78.75)

(D) (4, 750)

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

After 10 minutes, the Sandia Peak tram has traveled 12,000 feet. The whole ascent is 18,000 feet. How much farther must it travel to reach the top?

(A) 1,200 feet

(B) 5,000 feet

(C) 6,000 feet

(D) 12,000 feet

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

A second tram travels 900 feet per minute. If both trams are graphed on the same axes (minutes vs. feet), how do the graphs compare?

(A) The 900 ft/min line rises more steeply

(B) The two lines are identical

(C) The Sandia Peak line (1,200 ft/min) rises more steeply than the 900 ft/min line

(D) The 900 ft/min line goes downward

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Find a card you called a match and your partner did not. Pick one ordered pair from it and test it in cost = 45 × tickets — what does the point on the graph tell you that the table does not?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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